# ZIYAD MOHAMED ALBOGHDADY

Backend Engineer · C# | ASP.NET Core | Spring Boot | PostgreSQL | Docker
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## SUMMARY

Third-year Software Engineering student (full scholarship, 3.63 GPA, 4.00 most recent semester) building backend systems in C# and ASP.NET Core, currently a machine-learning intrusion-detection research intern at Sakarya University's CyberSense Laboratory. Rise Global Finalist and two-time hackathon winner.

## EDUCATION

**B.Sc. Software Engineering — Sakarya University** *2023 – Present*
Full merit scholarship · GPA 3.63 / 4.00 · Most recent semester 4.00 / 4.00 · AA in Data Structures & Algorithms, Database Systems, Operating Systems, Secure Software Development, Distributed Systems

## EXPERIENCE

**Research Intern, Machine Learning Intrusion Detection** — CyberSense Laboratory, Sakarya University *Aug 2026 – Present*
- Contributing to an ML-based DDoS detection system built as a native Snort 3 C++ inspector that classifies network flows from only the first 2 packets — XGBoost training, ONNX Runtime serving — over a self-labelled 1.19M-flow dataset extracted from raw pcap.
- Designed the feature-"lift" metric that proved a model scoring 99.96% accuracy relied on a dataset fingerprint rather than attack behaviour, isolating a victim TCP-window constant present in 99.74% of attack flows.
- Built host-disjoint train/test splits showing false-positive rate rose from a certified 0.037% to ~50% once the artefact was removed — documenting a measured limit instead of shipping an unreproducible result.

**Volunteer Translator (Arabic–English–Turkish)** — Turkish Ministry of National Defence, IDEF International Defence Exhibition *2025*
- Live technical and diplomatic interpretation for international delegations under confidentiality requirements.

## PROJECTS

**Task Management API** — .NET 9 Web API | PostgreSQL 16 | Docker Compose /dotnet-layered-architecture-task-api *2026*
- Containerised REST API with strict layered separation (Controllers → Services → Repositories), EF Core + Npgsql behind an ITasksRepository interface, and secrets loaded from .env and injected via compose.yaml.
- Correct REST semantics — 201 with Location header, PATCH partial update, 204 on delete — plus Swagger docs and a documented test proving data survives full container teardown via a named Docker volume.

**Car Rental Management System** — ASP.NET Core MVC | C# | EF Core | SQL Server /car-rental-management-system
- Full-stack platform across seven modules — cars, customers, employees, contracts, payments, licences, accounts — on a normalised schema with constraints, stored functions and triggers enforcing business rules in the database.
- ASP.NET Core Identity with role-based access, EF Core migrations, and shipped documentation (ERD, business rules, requirements specification).

**Academic Supervision API** — Spring Boot 4 | Java 21 | Spring Data JPA | PostgreSQL /academic-supervision
- Layered REST backend with DTO mapping, Jakarta Validation, custom exceptions and a global handler returning structured error responses; documented with Springdoc OpenAPI.

**ResQNet — Disaster Response & Crisis Coordination System** — Flutter | React | Firebase *2026*
- Built in a 4-person team over 24 hours: offline BLE / Wi-Fi Direct mesh SOS reporting, a transparent weighted triage algorithm chosen deliberately over opaque ML, and a rule-based volunteer-matching engine with explainable assignments. **2nd place — 24H Innovation Challenge, Sakarya University.**

**Axiom AI Carbon Coach** — React Native | Expo | TypeScript | Firebase | Gemini API /axiom *2026*
- Mobile carbon-footprint tracker with AI reduction advice, Google Calendar travel detection, camera product scanning, and a gamified leaderboard. **Best Design award — Düzce University hackathon.**

**Assistive Technology for Visually Impaired Users** — Embedded Systems | Computer Vision *2022*
- Smart navigation cane with audio obstacle alerts and an educational system for blind students, designed after three months volunteering with blind users at Resala Charity; extended with AI glasses doing object, face and currency recognition plus OCR text-to-speech. **Rise Global Finalist (top 500 of 131,000+) · 2nd place Egypt, 5th Middle East — IoT & AI Competition.**

## TECHNICAL SKILLS

**Languages:** C#, C++, Java, JavaScript, Python, SQL
**Backend:** ASP.NET Core (MVC, Web API, Minimal APIs), EF Core, Dapper, ASP.NET Identity, SignalR, Spring Boot, Spring Data JPA, REST API design, API versioning, middleware, dependency injection & service lifetimes, Swagger / OpenAPI
**Frontend:** React, JavaScript, Razor Views, Bootstrap, HTML5/CSS3
**Data & ML:** PostgreSQL, SQL Server, ERD design, normalisation, triggers & stored functions, XGBoost, feature engineering, data-leakage diagnosis, ROC/PR-AUC, Snort 3, pcap analysis
**Architecture:** OOP, SOLID, MVC, Repository Pattern, layered architecture, DTOs, async/await, authentication & authorization
**Tools:** Git, Docker & Docker Compose, Maven, Visual Studio, VS Code, Postman, Wireshark, Linux, Agile

## AWARDS, CERTIFICATIONS & LANGUAGES

**Rise Global Finalist** — top 500 of 131,000+ (2022) · **IoT & AI Competition** — 2nd place Egypt, 5th Middle East · 2nd place 24H Innovation Challenge (Sakarya Univ.) · Best Design, Düzce University Hackathon · Bronze Honours — International Youth Math Challenge, International Astronomy & Astrophysics Competition
FlyRank Backend AI Engineering track (remote), Jul 2026 – Present · GCI World 2026 — Data Science, Matsuo-Iwasawa Lab, University of Tokyo, Sep – Dec 2026 · Anthropic: Claude 101, Claude Code 101, AI Fluency · Rise Entrepreneurship Bootcamp, University of Chicago
**Languages:** Arabic (Native) · English C1 (IELTS Academic 7.0) · Turkish C1 · Russian A2 · French A1